Data communications (Lab 2): Amplitude Shift Keying (ASK)

Objective: To generate and plot an Amplitude Shift Keying (ASK) signal in MATLAB using a binary data sequence and a sinusoidal carrier wave.

Theory

Amplitude Shift Keying (ASK) is a digital modulation technique in which the amplitude of the carrier signal changes according to the binary input data.

- For binary **1**, the carrier signal is transmitted.
- For binary **0**, the carrier signal is suppressed or reduced.

Thus, the modulated output contains the carrier only during the periods where the input bit is 1.

Mathematically, $s(t) = b(t). \sin(2\pi f t)$

Where:

- $b(t)$ = binary data waveform
- $\sin(2\pi f t)$ =carrier signal
- $s(t)$ = ASK modulated signal

```
clear;                     % Clear all variables from workspace
clc;                       % Clear command window

b = [0 1 0 1 1 1 0];       % Input binary data sequence
n = length(b);             % Number of bits in the sequence

t = 0:.01:n;               % Time vector from 0 to n with step size 0.01

% Convert binary data into waveform (each bit expanded to 100 samples)
for i = 1:n
    bw(i*100:(i+1)*100) = b(i);    % Assign bit value over 100 samples
end

bw = bw(100:end);          % Adjust length of bw to match time vector

sint = sin(2*pi*t);        % Generate carrier sine wave (frequency = 1
                             Hz)

st = bw .* sint;           % Perform ASK modulation (element-wise
                             multiplication)

% -------- Plot Binary Data --------
subplot(3,1,1)             % First subplot (3 rows, 1 column, position 1)
plot(t,bw)                 % Plot binary waveform
grid on                    % Turn grid on
```

```
axis([0 n -2 +2])         % Set axis limits

% -------- Plot Carrier Signal --------
subplot(3,1,2)            % Second subplot
plot(t,sint)              % Plot sine wave (carrier)
grid on
axis([0 n -2 +2])

% -------- Plot ASK Signal --------
subplot(3,1,3)            % Third subplot
plot(t,st)                % Plot ASK modulated signal
grid on
axis([0 n -2 +2])
```

Result

The program produces three plots:

1. **Binary data signal**
2. **Carrier signal**
3. **ASK modulated signal**

In the final output:

- Carrier exists during bit **1**
- No carrier during bit **0**

This confirms proper ASK modulation.